



WiFi Signal Strength Monitoring System

Using NodeMCU + I²C LCD + LED Indicators + Buzzer Alert



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Area of Work & Motivation

Why this project matters



Area of Work



IoT & Embedded Systems

NodeMCU microcontroller programming



Wireless Networking

RSSI measurement & signal analysis



Real-Time Sensing

Continuous signal monitoring loop



Hardware Integration

LCD, LEDs, Buzzer via GPIO pins



Motivation

- WiFi is the backbone of modern connectivity yet its quality is invisible to users
- Dead zones & signal drops cause frustration in homes, offices, and campuses
- Existing tools are either expensive or require complex software
- A low-cost hardware solution with visual + audio feedback fills this gap
- NodeMCU's built-in WiFi makes it the perfect embedded platform for this task

Problem Statement

The core challenge this project addresses

There is no simple, affordable, real-time system to measure, display, and alert users about WiFi signal strength variations across physical locations — making it difficult to identify dead zones and optimize network placement.

Invisible Dead Zones



Users cannot see where WiFi signal drops below usable levels in their environment

No Real-Time Feedback



Standard devices do not display RSSI continuously or alert users to poor signal

Costly Existing Solutions



Professional WiFi analyzers are expensive and overkill for simple coverage mapping

IoT Device Reliability



IoT sensors in weak zones fail silently; there is no mechanism to trigger movement

Proposed Solution

How we address the problem

A portable, low-cost NodeMCU-based system that continuously reads WiFi RSSI, classifies signal strength, and provides real-time visual (LCD + LEDs) and audio (buzzer) feedback.



NodeMCU Controller

- Built-in WiFi module
- Reads RSSI every 300ms
- Classifies: Strong / Medium / Weak
- Connects to any hotspot



Visual Output

- 16×2 I²C LCD shows RSSI in dBm
- 3 LEDs mimic phone signal bars
- Strong → 3 ON, Medium → 2, Weak → 1
- LCD shows 'Weak Zone — Move Closer'



Audio Feedback

- Active buzzer for alerts
- Medium → slow beep (200/800ms)
- Weak → continuous loud alert
- Strong → buzzer stays OFF

Summary of Work Done

Key milestones accomplished this semester

Hardware Assembly

1



NodeMCU wired to 16x2 I²C LCD (D1/D2), 3 LEDs via 220Ω resistors (D5/D6/D7), and active buzzer (D0) on breadboard

WiFi Connectivity

2



Programmed NodeMCU to connect to a WPA2 hotspot; displays assigned IP address on LCD after successful connection

Network Scanner

3



Implemented WiFi.scanNetworks() to list all nearby SSIDs with RSSI values and identify the strongest available network

RSSI Classification Engine

4



Built real-time signal classifier: Strong (>-65 dBm), Medium (-65 to -75 dBm), Weak (<-75 dBm) with 300ms polling loop

LED + LCD Output System

5



LCD prints RSSI and status text; LED bar (3/2/1) mirrors phone signal bars; 'Weak Zone — Move Closer' shown on weak signal

Buzzer Alert System

6



Coded two alert patterns: slow beep (200ms ON / 800ms OFF) for medium and continuous tone for weak signal conditions

Results

Observed system behaviour during testing

STRONG

RSSI > -65 dBm

```
RSSI: -50 dBm  
Strong [||||]
```

LEDs: **3 LEDs ON**

Buzzer: **Buzzer OFF**

Ideal connectivity

MEDIUM

-65 to -75 dBm

```
RSSI: -70 dBm  
Medium [|| ]
```

LEDs: **2 LEDs ON**

Buzzer: **Slow Beep (200ms/800ms)**

Move slightly closer

WEAK

RSSI < -75 dBm

```
RSSI: -82 dBm  
Weak Zone
```

LEDs: **1 LED ON**

Buzzer: **Continuous Alert**

Change location!

Network Scanner also successfully detected all nearby SSIDs, ranked by RSSI, and identified the strongest available network.

Importance of Results

Why these outcomes matter

Proves Concept Works End-to-End



The system correctly classifies all three signal tiers and triggers the right output for each — validating the full hardware-software pipeline

Real-Time Response at 300ms



Sub-second RSSI updates mean users receive immediate feedback as they move through a space — critical for practical coverage mapping

Enables Physical Coverage Mapping



By walking through rooms and reading RSSI, users can accurately identify dead zones and reposition routers for optimal coverage

Dual Feedback Reduces Errors



Combining LED bars + buzzer audio means the device is usable even when the LCD is not directly visible — improving real-world utility

Network Scanner Adds Practical Value



Scanning all nearby SSIDs and ranking by RSSI lets users compare networks and select the strongest one — beyond just monitoring one connection

Foundation for IoT Deployment



Results demonstrate NodeMCU can serve as a signal-aware IoT node, with clear upgrade paths to cloud logging and mesh networking

Future Work

Planned enhancements and extensions

Mobile App Dashboard

1



Transmit RSSI data over WiFi to a companion smartphone app for remote real-time monitoring and heatmap visualization

Auto Network Switching

2



Automatically disconnect and reconnect to the strongest available network when signal drops below a configurable threshold

Cloud Data Logging

3



Push timestamped RSSI readings to a cloud database (Firebase / ThingSpeak) for historical trend analysis and reporting

2D Coverage Heatmap

4



Log GPS or room coordinates alongside RSSI to generate a visual heatmap of signal distribution across a building

Mesh Network Support

5



Extend to multi-node setup where several NodeMCU units collaborate to map an entire floor and relay data to a central hub

OLED / TFT Display Upgrade

6



Replace 16x2 LCD with a colour OLED / TFT screen to display real-time signal graphs, SSID list, and battery level simultaneously

Challenges & What Could Not Be Done

Honest reflections on limitations



Challenges Faced

● I²C Address Conflict

LCD at 0x27 conflicted with other I²C devices; resolved by scanning and hardcoding the correct address

● RSSI Fluctuation

RSSI values fluctuated ± 3 –5 dBm even when stationary; solved with a rolling average buffer

● 5V LCD Power Issue

LCD required 5V but NodeMCU runs on 3.3V; resolved by powering LCD via Arduino UNO's 5V rail

● Buzzer Timing vs Loop

Non-blocking buzzer beep patterns interfered with the 300ms polling loop; required careful delay management

● WiFi Reconnect Delay

When hotspot was lost, reconnection took 5–10s; added a status check and reconnect routine in the loop



What Could Not Be Done

■ Mobile App Integration

Planned but HTTP server on NodeMCU was unstable under continuous RSSI polling; deferred to future work

■ Persistent Data Logging

No SD card or cloud logging implemented — all data is transient and lost on power cycle

■ 2D Coverage Heatmap

Spatial RSSI mapping was conceptualized but GPS/coordinate logging was not integrated this semester

■ Battery-Powered Operation

Prototype runs on USB only; a regulated LiPo battery pack was not implemented due to time constraints

■ Multi-Band WiFi Support

NodeMCU only supports 2.4 GHz; 5 GHz RSSI measurement was out of scope for this hardware